

Nanomaterials

Introduction:

Nanomaterials are cornerstones of nanoscience and nanotechnology. Nanostructure science and technology is a broad and interdisciplinary area of research and development activity that has been growing explosively worldwide in the past few years. It has the potential to revolutionize how materials and products are created and the range and nature of functionalities that can be accessed. It is already having a significant commercial impact, which will assuredly increase in the future.

Nanoscience: The study of phenomena and materials at the atomic, molecular and macromolecular scales, where properties differ significantly from those at the larger scale

Nanotechnology: Design, characterization, production and application of structures, devices and systems by controlling morphology (size, shape and distribution) at the nano-scale.

Q What are nanomaterials?

Nanoscale materials are defined as a set of substances where at least one dimension is less than approximately 100 nanometers. A nanometer is one-millionth of a millimetre- approximately 100,000 times smaller than the diameter of a human hair. Nanomaterials are of interest because unique optical, magnetic, electrical, and other properties emerge at this scale. These emergent properties have the potential for great impacts in electronics, medicine, and other fields.

Q How are nanomaterials synthesised?

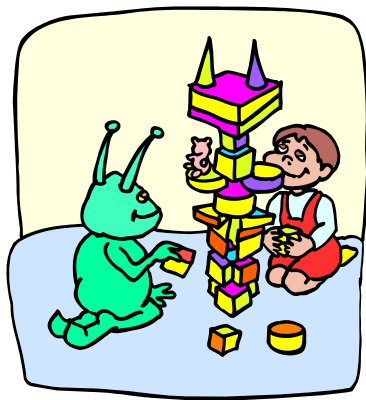
Nanomaterials are made by two generalized processes

(i). Top-down: Start with the big chunk and cut away material to make what you want. It begins with bulk materials that are subsequently reduced into nanomaterials.

Ex: Inert gas condensation, Ion beam technique, Laser ablation, etc

(ii). Bottom-up: Building what you want by assembling it from small prefabricated units molecules react under chemical or physical circumstances to form nanomaterials. such as atoms and molecules. It begins with atoms and molecules/. These atoms or

Ex: Hydrothermal, Sol-gel method, Co-precipitation and Combustion synthesis



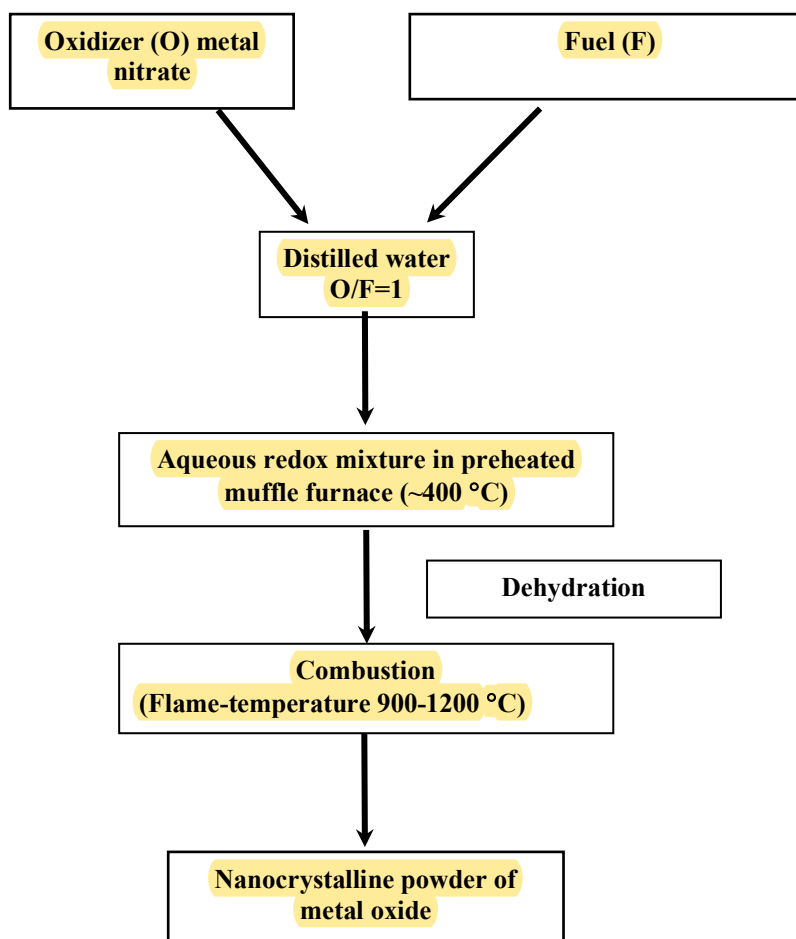
Bottom-up



Top-down

Q. What is solution combustion synthesis? Explain the synthesis of nano ZnO particles using the solution combustion method.

The solution combustion process is a self-propagating high-temperature synthesis in which an exothermic reaction takes place accompanied by the evolution of gases



Flow – Chart for synthesis of nanoparticles by solution combustion method

The solution combustion method is a wet-chemical method, which has been proven to be an excellent technique for preparing nanocrystalline materials due to its short

processing time and low processing temperature. This technique is based on exothermic redox reactions that undergo self-sustaining combustion. Mixtures of oxidizer (usually metal nitrates) and a fuel (an organic compound containing carbon and hydrogen; e.g. urea, citric acid or glycine as fuel) undergo spontaneous combustion under heating and the chemical energy from the exothermic reaction heats the precursor mixture to high temperatures. Such a high temperature leads to the formation and crystallization of nanomaterials.).

Advantages of the solution combustion method:

Among wet chemical routes, the low-temperature ***solution combustion process*** is an attractive one and finds several advantages-

- i. The synthesis temperature (300°C) employed is lower than those currently used in conventional routes.
- ii. Simple and energetically attractive process.
- iii. Ability to dope desired amounts of impurity ions with better homogeneity.
- iv. The powders are voluminous, foamy, sinteractive and in nanoscale having large surface area.
- v. short reaction time (a few seconds to 5 minutes)
- vi. Chemical homogeneity

Example: Synthesis of ZnO nanoparticles by solution combustion method: Refer to lab manual, page -24

Q. Explain the synthesis of nanoparticles using the hydrothermal method

It is a solution-based method to prepare a wide range of nanomaterials having diverse shapes and sizes. It is a method to produce different chemical compounds and materials using closed systems in aqueous solutions at temperatures above 100°C and pressures above 1 atm.

Principle: It is based on the principle of dissolution and recrystallizing substances from high-temperature aqueous solutions at high vapour pressure. The crystal growth is performed in an apparatus consisting of a steel pressure vessel called an autoclave, in which a nutrient (precursors) is supplied along with water. A gradient of temperature is maintained at the opposite ends of the growth chamber so that the hotter end dissolves the nutrient and the cooler end causes seeds to take additional growth.

Procedure: In typical hydrothermal synthesis, a stoichiometric amount of precursors and ~20-25 ml of double distilled water is taken in a Teflon liner. In a few cases to

maintain the basic pH alkali like NaOH or KOH are also added. Then the mixture is stirred well using a magnetic stirrer to ensure complete homogenous mixing until all the contents are evenly distributed. Later, the Teflon liner is kept inside the autoclave and sealed firmly. The complete autoclave assembly is then kept in a Hot air oven at a desired temperature (generally $< 150\text{ }^{\circ}\text{C}$) and the reaction is carried for the desired time (Depending on the material time varies between 2 and 24 h). Then, after the reaction, the product is washed with water and ethanol to remove impurities and the products are dried at $60\text{ }^{\circ}\text{C}$ for 3 h in a dustproof hot air oven to get nanocrystalline materials.

For example: The preparation of BiFeO_4 by hydrothermal-Equimolar mixtures of bismuth nitrate ($\text{Bi}(\text{NO}_3)_3 \cdot 5\text{H}_2\text{O}$) and iron nitrate ($\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$) are dissolved in distilled water. Add 8 M sodium hydroxide (NaOH) stir for 30 min and pour into an autoclave. The hydrothermal treatment is conducted at 180°C for 12 h. The produced powder is collected at the bottom of the autoclave after cooling to room temperature. The product is washed several times by repeated cycles of centrifugation in distilled water and ethanol. The obtained powder is heated at 80°C for 1 hr.

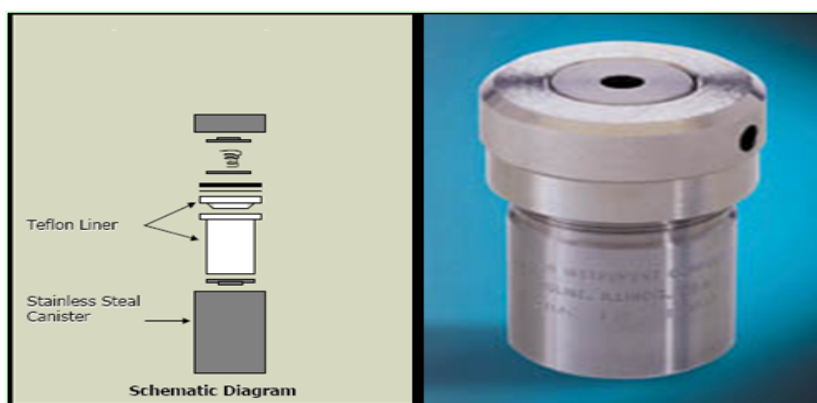


Fig. Autoclave for synthesis of nanomaterials by hydrothermal method

Q-Mention the various techniques used in the characterization of nanopowder sample

The various characterization techniques used are—(i) Powder X-ray diffraction (ii) Scanning electron microscopy (SEM) (iii) Transmission electron microscopy (TEM) (iv) Infrared spectroscopy (v) Ultraviolet-Visible spectroscopy

What are the applications of nanomaterials?

Nanomaterials possess unique optical, electronic and magnetic properties that make them unique and widely applicable in various fields.

Application (field)	Examples
Medicine	Diagnostics, Drug delivery, Tissue engineering, Cryonics
Information and communication	Memory storage, Novel semiconductor devices, electronic devices, Displays, Quantum computers
Heavy Industry	Aerospace, Catalysis, Catalysis, Construction Vehicle manufacturers
Consumer goods	Foods, Household, Optics, Textiles, Cosmetics, Sports
Environment	Water treatment: The removal of colour, heavy metal, fluoride, etc Nanomaterials have better photo catalytic activity than their bulk counterparts due to very high surface area. Nano photo catalysts like ZnO, TiO ₂ and CuO are used to degrade organic dyes and other impurities from waste industrial waste water.

Unit-4 Electrochemical Sensors

Analytical chemistry (techniques) deals with chemical analysis which involves qualitative (identification) and quantitative (estimation or determination) analysis of the percentage composition in a sample. Analytical chemist is an integral part of the R&D section of any industry.

Quantitative analysis

1. Classical methods of analysis or Chemical methods of analysis: Chemical methods of analysis involve gravimetric and volumetric analysis. In gravimetry, a known amount of analyte is precipitated by the addition of an organic reagent and estimation by weight. Volumetry or titrating analysis involves the determination of the volume of a solution of known concentration which is required to react quantitatively with a measured volume of a solution of the substance to be determined.

Limitations of classical methods

i. Time-consuming procedures ii. Lack of versatility iii. Lack of accuracy when a small quantity of substances is involved

2. Modern method of analysis or Instrumental methods of analysis: In an instrumental method of analysis, the physical property of a substance is measured to determine its chemical composition. An instrument used for chemical analysis converts the property of the substance under investigation into a form that can be readily measured. The measured quantity is related to the quantity or quality of the sample.

Ex: Colorimeter measures the amount of radiation absorbed by a coloured solution by converting the light radiation into an electrical signal. Further, the amount of light radiation absorbed is a function of concentration of the solution and the wavelength at which the absorption takes place, is the characteristic of the substance present.

Advantages

1. Faster than chemical method 2. Applicable at concentrations too small for determination by classical methods. 3. Find wide applications in industries 4. The analytical process can be automated 5. Small quantities of the analyte are sufficient. 6. Gives accurate results.

Disadvantages

1. Instruments are expensive 2. An initial or continuous calibration is required using a sample of material of known composition 3. The concentration range is limited 4. Specialized training is needed for the operation of instruments

Types of Instrumental Methods of Analysis:

i. **Electrical methods (Electro analytical):** The electrical method of analysis involves the measurement of current, voltage or resistance concerning the concentration of a certain species in solution. For example, potentiometry involves the measurement of the potential of an electrode in equilibrium with an ion to be determined.

ii. **Optical methods (Spectro analytical):** Optical methods are based on, how the sample acts towards the electromagnetic radiation. For example, in colorimetry, the intensity of the color of the solution is a measure of the concentration of the substance. The intensities are determined visually or with the help of a colorimeter.

Sensors:

"The sensor is a device that interacts with a specific chemical or biological species (analyte), detects it and measures its physical/ chemical information into a measurable electrical signal proportional to its quantity".

Physical/Chemical phenomena such as Temperature, pressure, motion, light, or sound/composition, concentration, chemical activity, partial pressure etc.

The information may originate from a chemical reaction of the analyte or from a physical property.

Sensors play a key role in

- Environmental monitoring
- Industrial process monitoring
- Gas composition analysis
- Medicinal field for disease diagnosis and health monitoring
- National defence and public security
- Agricultural area for soil monitoring and on-site pesticides, herbicides monitoring etc.

The following are the important properties of the sensor.

- Small size
- Satisfactory sensitivity
- Larger dynamic range
- Low cost
- Quick response
- Accuracy of results
- Easy to realize automatic measurement
- Online or in-situ and continuous detection.

Classification of Chemical sensor

Chemical sensors may be classified according to the working principle of a transducer into the following types.

- ✓ Optical chemical sensor
- ✓ Electrochemical devices
- ✓ Electrical devices
- ✓ Mass sensitive devices
- ✓ Magnetic devices
- ✓ Thermometric devices

Electrochemical sensors:

Electrochemical sensors are devices that detect and measure the concentration of a target analyte by converting the effect of the electrochemical reaction between the analyte and electrode surface into an electrical signal.

Analytical information of the sample is extracted from the measurement of electrical parameters such as potential, current or conductance.

In electrochemical sensors, the electrode or a set of electrodes is used as a transducer element. Hence these sensors are often called Electrochemical cells.

Depending on the mode of signal transduction, electrochemical sensors can be classified into following types

Potentiometric sensor (measure voltage)

Amperometric sensor (measure current)

Conductimetric sensor (measure resistance or conductivity)

Applications of sensors

1. Electrochemical sensors used for the detection of blood glucose
2. Electrochemical sensors are used for pH measurements
3. They are used in soil parameter analysis and in agriculture applications to detect pesticides.
4. Used in the detection of hydrocarbon pollutants
5. Optical sensors are used to measure the concentration of coloured analyte.
6. The use of electrochemical techniques for detecting enzymes and hormones to monitor cancer.

Q- What is potentiometry? Explain its principle/theory of potentiometer.

The procedure of using measurement of e.m.f to determine the concentration of ionic species in solution is referred to as potentiometry.

The relation between electrode potential and metal ion concentration is given by the Nernst equation.

$$E = E^{\circ} + \frac{0.0591}{n} \log [M^{n+}]$$

It can be seen from the equation that the potential of an electrode E depends upon the concentration of the ion M^{n+} to which it is reversible.

The principle involved in potentiometric titration is the measure of emf between two electrodes, an indicator electrode, the potential of which is a function of the concentration of the ion to be determined and a reference electrode standard calomel electrode (SCE) of constant potential. In these titrations, the measurements of emf are made while the titration is in progress. The equivalence point of the reaction is revealed by a sudden change in potential in the plot value against the volume of the titrant.

Instrumentation:

A potential consists of a reference electrode, an indicator electrode and a device for measuring the potential. The indicator electrode responds rapidly to the changes in the potential due to the concentration changes of the analyte.

A known volume of the analyte is taken in a beaker and its potential is determined by connecting the assembly to a potentiometer. The titrant is added in increments of 0.5 ml and the potential is measured each time. Close to the equivalence point, there is a sharp increase in the potential ... At this point, small readings are taken beyond the endpoint. The endpoint is determined by plotting the change in potential against the volume of titrant.

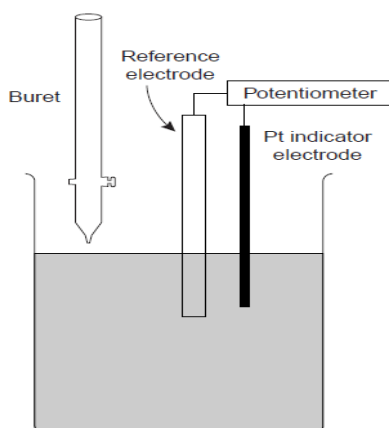


Fig. Potentiometer

Q Explain the determination of Iron using a potentiometer

Refer manual – Page 7

Optical Sensors -Colorimetry:

Optical Sensors: “Optical sensors are electronic components designed to detect and convert incident light rays interacted with analyte into measurable electrical signals for detecting and measuring physical or chemical properties of an analyte “.

- ❖ Optical sensors are based on the transduction of the interaction of electromagnetic radiation with chemical species.
- ❖ Commonly electromagnetic radiations in the UV- visible-IR domains are used for analytical purposes.
- ❖ In optical sensors, the optical signal arises from the interaction of the analyte with incident radiation. The interaction could result in absorption, emission, reflection and scattering of light.
- ❖ The type of interaction depends on the wavelength of the electromagnetic radiation and the structure of the molecules of an analyte.
- ❖ The intensity of the radiation emitting or transmitting from the analyte carries information on the concentration of the analyte.
- ❖ It is measured by the optoelectronic instrument.

Absorption-based optical sensors-Colorimetric Sensors:

Colorimetric sensors are simple types of optical sensors that use incident light whose wavelength falls in the visible region (400-800nm) of the spectrum and detect and determine the concentration of coloured chemical species of analyte solutions.

- ❖ The wavelength of Ultraviolet radiation, the visible region lies between 8000-4000Å and the UV region lies between 4000–2000Å. Colorimetry is concerned with the visible region of the spectrum.
- ❖ Colorimetry is used to determine the concentration of compounds in a solution that are themselves colored or which give color when mixed with a suitable reagent. The intensity of the color of the solution is a measure of the concentration of the substance. The intensities are determined visually or with the help of a colorimeter.
- ❖ **Colorimetry:** Chemical analysis through measurements of absorption of light radiation in the visible region of the spectrum concerning a known concentration of the substance.
- ❖ **Colorimeter:** An instrument used for measuring absorption in the visible region.
- ❖ **Beer's Law:** the intensity transmitted light decreases exponentially as the concentration of the medium increases arithmetically $I_t = I_0 e^{-kc}$ (1)
- ❖ **Lambert's- Law:** the intensity transmitted light decreases exponentially as the thickness absorbing medium increases arithmetically

$$I_t = I_0 e^{-kt} \text{(2)}$$

From equation (1) and (2)

$$I_t = I_0 e^{-kct}$$

$$\log(I_0/I_t) = \epsilon ct \text{ (Beer-Lambert law)}$$

$$A = \epsilon ct$$

A = absorbance

c = concentration of the solution

T = path length

ϵ = molar absorption which is a constant at a given wavelength.

If the path length is kept constant then A is proportional to c. The absorbance of the solution is measured at an optimum wavelength.

Explain its principle/theory of colorimetry.

Principle/Theory: When an analyte solution interacts with incident light of a suitable wavelength, a certain quantity of light is absorbed by the analyte solution, and a chemical or physical change occurs with incident light. A transducer such as a photodetector converts the intensity of light absorbed by the analyte into an electrical signal.

A monochromatic light whose wavelength falls in the visible region of the spectrum is made to pass through an analyte solution where a certain quantity of incident light is absorbed and the remaining amount of light is transmitted. The concentration of coloured chemical species in the analyte solution is determined by measuring the absorbance or transmittance of light of a particular wavelength. This type of analysis is governed by Beer- lamberts law $A = E Ct'$

Instrumentation:

The photoelectric colorimeter consists of the following parts

- 1. Light source:** The wavelength range of visible light lies between 4000 – 8000Å. In this region, a tungsten filament lamp is most widely used.
- 2. Converging lens:** Light after passing through slits falls on the converging lens which gives a parallel beam of light.
- 3. Wavelength selector or Filter:** A source generally emits continuous spectra. Therefore, a device is required to select a narrow band from a wavelength of the continuous spectra. For this selection, filters (or) monochromators or both are used.
- 4. Slit:** It is adjustable which allows only a beam of light to pass through. It prevents unwanted or stray light.
- 5. Sample holder:** The cells holding the sample should be transparent to the wavelength region being recorded. For the visible region, generally, cells are made of

coloured corrected fused glass. The thickness (internal distance) of the cell is generally 1 cm.

6. **Photodetector:** To detect radiation, the three types of photosensitive devices i.e., photovoltaic cells, phototubes and photomultiplier tubes are used.

7. **Read-out device.**

Block diagram of a colorimeter

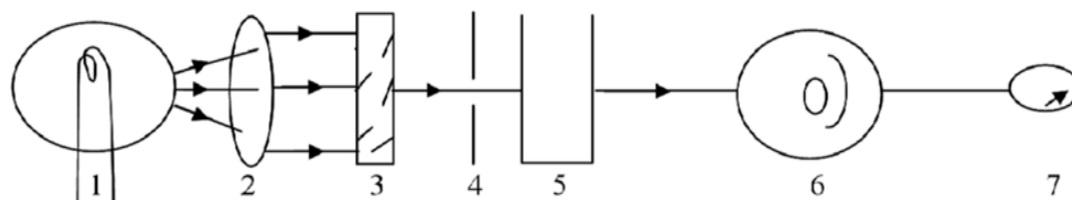


Figure. Components of Colorimeter

Working: Light from the tungsten lamp, after passing through the filter is allowed to fall on the solution taken in the sample cell. First, a blank solution is taken in the sample cell and placed in the path of a light beam. Its absorbance is adjusted to zero (or transmittance placed in the path of the light beam and the quantity of light absorbed is measured as its absorbance (or transmittance)).

Q Explain the determination of copper calorimetrically

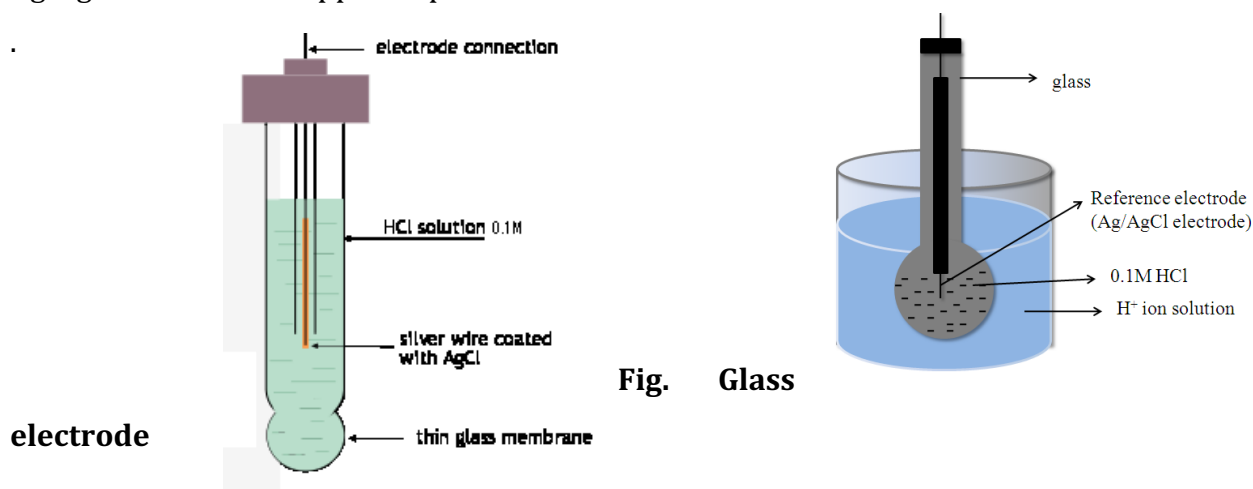
Refer to manual–Page number 3

Glass electrode- its application in the determination of the pH of beverages.

A glass electrode is an example of an ion-selective electrode, which selectively responds to specific ions in a mixture and the potential developed at the electrode is a function of the concentration of that ion in the solution. Ion-selective electrodes make use of a kind of membrane which is sensitive to a particular chemical species. Therefore, these electrodes are also known as membrane electrodes. The electrode generally consists of a thin membrane which is capable of exchanging the specific ion with the solution it is in contact.

Principle of Glass electrode: If solutions of different pH are separated by a thin membrane, the potential develops across the membrane. The magnitude of potential depends on the difference in pH of the two solutions. If the pH of one of these is held constant, the observed glass electrode potential is a linear function of the pH of the second solution.

Construction: A glass electrode is an ion-selective reference electrode. It consists of a thin-walled glass bulb of a special type with high electrical conductivity and with low melting point. A portion of the bulb is filled with a solution of constant pH (0.1 M HCl). Ag-AgCl electrode is dipped to provide electrical contact.



Notation of the electrode potential: Glass electrode can be represented as

Ag/AgCl (s)/0.1 M HCl / glass

Glass electrode Potential $E_G = E_G^0 - 0.0591 \text{ pH}$

E_G^0 is specific to a particular glass electrode.

Q- Explain the determination of the pH of an unknown solution using a glass electrode.

The glass electrode is used as the internal reference electrode for determining the pH of solution especially colored solution containing oxidising or reducing agents. Calomel electrode is used as the secondary reference electrode. To determine the pH of the given solution, the glass electrode is placed in the test solution and this half-cell is coupled with a saturated calomel electrode (Fig). The glass electrode acts as an anode calomel electrode behaves as a cathode. The *emf* of the cell is determined potentiometrically (pH meter)

$$\begin{aligned} E_{\text{cell}} &= E_{\text{cathode}} - E_{\text{anode}} \\ &= E_{\text{glass}} - E_{\text{SCE}} \\ &= [E_G^0 - 0.0591 \text{ pH}] - E_{\text{SCE}} \text{ [since } E_G = E_G^0 - 0.0591 \text{ pH}] \end{aligned}$$

$$0.0591 \text{ pH} = [E_G^0 - E_{\text{cell}} - E_{\text{SCE}}]$$

$$\text{Or } \text{pH} = [E_G^0 - E_{\text{cell}} - 0.241]/0.0591$$

[Since Electrode potential of saturated calomel is 0.241V]

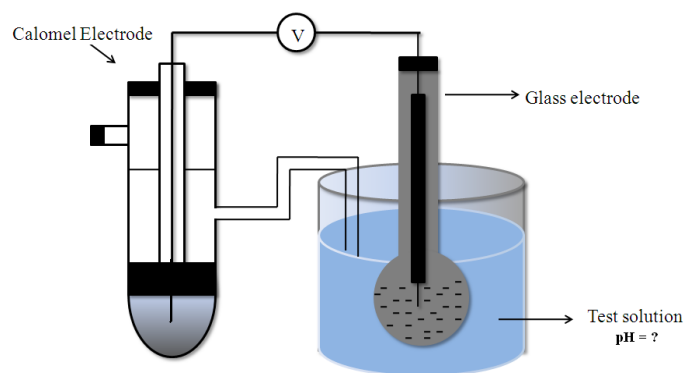


Fig. Determination of pH of an unknown solution